

Product Requirement Document (PRD) - QR Code-Based Attendance System

Field	Details
Feature Name	QR Code Attendee Check-In & Digital Member Pass
Status	Implemented
Target Audience	Meetup Attendees & Community Admins
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1. Executive Summary & Objective

Managing attendance at large community events is often a slow, manual process prone to check-in queues and duplicate entries. The **QR Code Attendance System** replaces manual check-ins with a scan-based verification mechanism. Every registered user has a unique, persistent digital pass containing their user identity. Event organizers can scan this pass using their device's webcam to instantly check attendees in, providing real-time database updates, visual confirmation, and pleasant sound indicators.

2. User Journeys

A. The Attendee Check-In Experience

1. **Registration:** User registers for a meetup by submitting answers to the onboarding questions.
2. **Accessing Pass:** Once registered, the user accesses their persistent **Digital Member Pass** from the registration card on the meetup details page.
3. **Check-In:** At the event, the attendee displays the QR code on their mobile device to the organizer's camera.

B. The Admin Curation Experience

1. **Opening Scanner:** Admin views the meetup page and clicks the  **"Scan QR Code"** button.
2. **Scanning:** Admin grants camera access, selects the preferred camera input, and scans incoming attendees' screens.
3. **Instant Action:** The scanner automatically validates the pass, plays a chime on success (or buzzer on error), displays the attendee's name/picture on screen, registers check-in, and resumes scanning automatically after 2 seconds.

3. Functional Requirements

A. Unique Digital Member Pass

- Every user has a unique, persistent digital pass structured as a cosmic glassmorphism card.
- **Card UI Elements:**
 - Converge branding with gradient styling.
 - Attendee Name & Professional Meta (e.g. "Designer @ Pixels").
 - Perforated ticket boundary (dashed line with round circular punch notches on both sides).
 - Dynamic QR Code containing the validation payload:

```
{"userId": 4, "name": "Attendee Name", "type": "converge-user-p
```

- Simulated card barcode at the bottom for high-fidelity ticket styling.

B. Dynamic QR Code Generator

- QR codes are generated client-side dynamically using the free, secure public API: <https://api.qrserver.com/v1/create-qr-code/?size=220x220&data=...>
- The payload is URI-encoded to ensure correct parsing.

C. Live Webcam Scanner (Admin Only)

- Admin-only access. The camera interface is embedded in a modal.
- Includes a drop-down select element to switch between multiple video capture inputs (e.g., front vs. back cameras on a phone).
- Features a **laser beam scanning animation** (horizontal crimson bar moving vertically in loop) to signal scanner activity.

D. Audio-Visual Verification & Feedback

- **Feedback Overlay:** An overlay appears directly inside the scanner window during scanning:
 - **Valid Check-In:** Green cosmic glow card, displaying "✓
Checked In" along with the attendee's name.
 - **Invalid/Error Scan:** Red cosmic glow card, displaying "✗
Failed" along with the error reason (e.g., "Not registered for this meetup", "Already checked in").
- **Browser-Synthesized Sounds:** Uses the HTML5 Web Audio API to trigger instant sound indicators without relying on external static files:
 - **Success Sound:** Double happy chime (D5 followed by A5).
 - **Error Sound:** Sawtooth wave buzzer chime.
- **Automated Resume:** The overlay displays for exactly 2 seconds before automatically closing to resume scanning for the next attendee (hands-free operation).

- **Background Synchronization:** While scanning, the attendee listings, registration metrics, and checked-in counts on the page update dynamically in the background without refreshing the page.

4. Technical Architecture

A. Database Integrations

Reuses the existing sqlite schema `registrations` table's check-in capabilities:

```
CREATE TABLE IF NOT EXISTS registrations (  
  id          INTEGER PRIMARY KEY AUTOINCREMENT,  
  user_id     INTEGER NOT NULL,  
  meetup_id   INTEGER NOT NULL,  
  status      TEXT NOT NULL DEFAULT 'registered', -- checked_in |  
  checkin_time DATETIME,  
  ...  
);
```

B. API Reference

Method	Endpoint	Auth	Description
POST	<code>/api/meetups/:id/admin-checkin</code>	Admin Only	Validates user ID, checks registration status, marks check-in, and returns attendee name

C. Library Integration

- Incorporates the `html5-qrcode` library for robust camera handling, barcode decoding, and stream management.
- Script source: `https://unpkg.com/html5-qrcode` loaded globally.

- Instantiated via custom JS controls in `meetup.js` to ensure the webcam process can be cleanly stopped when the modal is closed, preventing camera privacy light leaks.